AI-Powered Customer Experience

LLMs as business components, and how to tell if a model is too good to be true

AI in Business · Week 4

Reading: Huyen, AI Engineering, ch. 1–2 and 5

Last week’s model, and today’s plan

0.96

AUC of a churn model a colleague sends you. Ours from week 3 scored 0.84.

Promote the colleague, or check the work?

1. **Too good to be true: leakage**
   The time test for every feature
2. **LLMs as business components**
   Tokens, cost, hallucination, structured output
3. **AI in customer service**
   Automate, assist, analyse; the evidence
4. **Can we trust it?**
   Evaluation and liability

One extra column, a much “better” model

Week 3 model

0.84

AUC · accuracy 81%

+ calls_to_cancellation_line

0.96

AUC · accuracy 92%

The new column

Calls to the cancellations line in the last 30 days.

80% of churners called it. Only 8% of stayers did.

Customers call that line after they decide to leave.

Leakage: a feature that carries information about the outcome which you would **not** have at the moment you make the decision. The model looks great in testing and fails in real use.

Activity 1: Spot the leak

A. Bank: will this loan default?

Number of calls from the **collections department** in the last 6 months

B. Hotel: will this booking cancel?

Assigned room type differs from the room type booked

C. Retail: will this customer use the coupon?

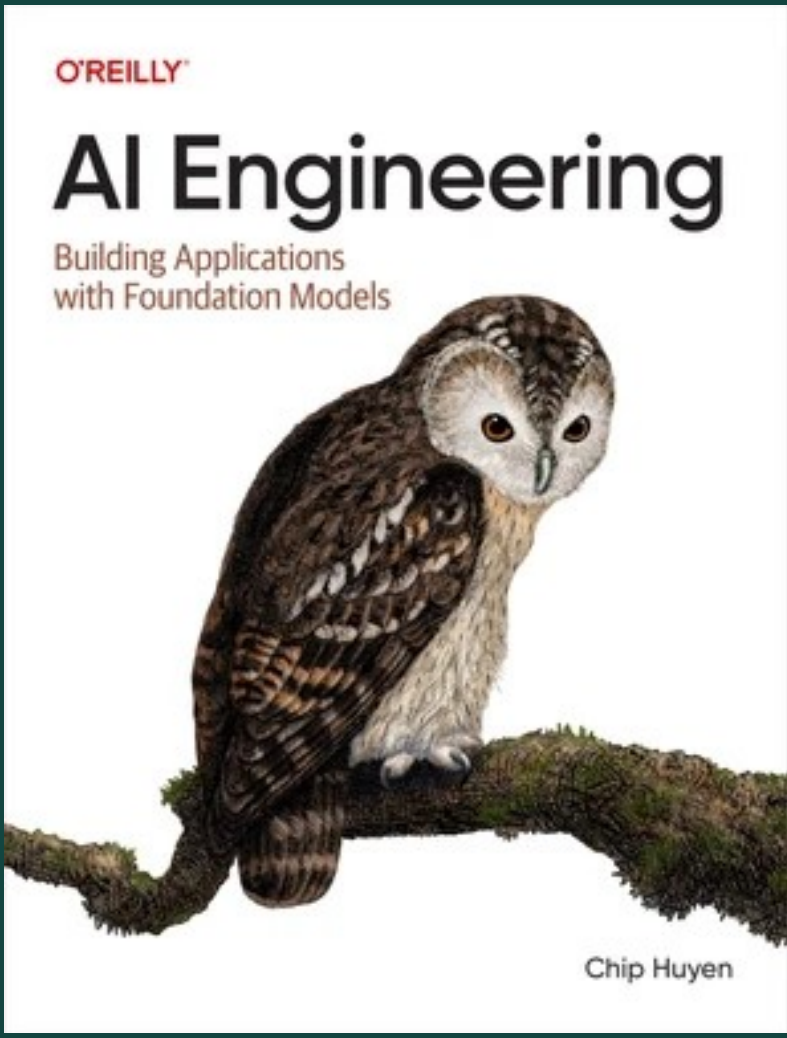
Total spent in the 30 days after the coupon was sent

D. Telco: will this customer churn?

Number of support tickets in the last 3 months

LLMs as business components

Reading Assignment: Huyen, AI Engineering, ch. 1–2 and 5



What an LLM actually does: guess the next word

"I have called you three times about my wrong …"

Next word	Model's guess
bill	62%
invoice	18%
address	7%
… thousands more	13%

Illustrative numbers. It picks a likely word, adds it, and repeats.

Plausible, not checked

It writes what sounds right. Nothing inside checks facts.

Doesn't know your company

Your prices and policies are not in it, unless you put them in the prompt.

Can answer differently each time

For business tasks, turn the randomness down (temperature 0).

Tokens: the unit you pay for

≈ ¾ word

1 token in English. Albanian text usually needs more tokens.

$0.30

per million tokens **in** (what you send)

$2.50

per million tokens **out** (what it writes)

MegaTelCo: 10,000 complaint emails a month

In: 10,000 × 500 tokens = 5 million × $0.30 = **$1.50**

Out: 10,000 × 50 tokens = 0.5 million × $2.50 = **$1.25**

AI total: **≈ $3 a month**

By hand: 1 minute each = 167 hours ≈ **€1,300** at €8/hour

The API bill is rarely the problem. The errors are.

Prices: Gemini 3.5 Flash-Lite, paid tier, Google AI pricing page

Hallucination: fluent is not the same as true

Mata v. Avianca, New York, 2023

Lawyers filed a brief citing six court cases found with ChatGPT. None of the cases existed. The judge fined them $5,000.

The tool did not lie on purpose. It produced what a legal citation *looks like*.

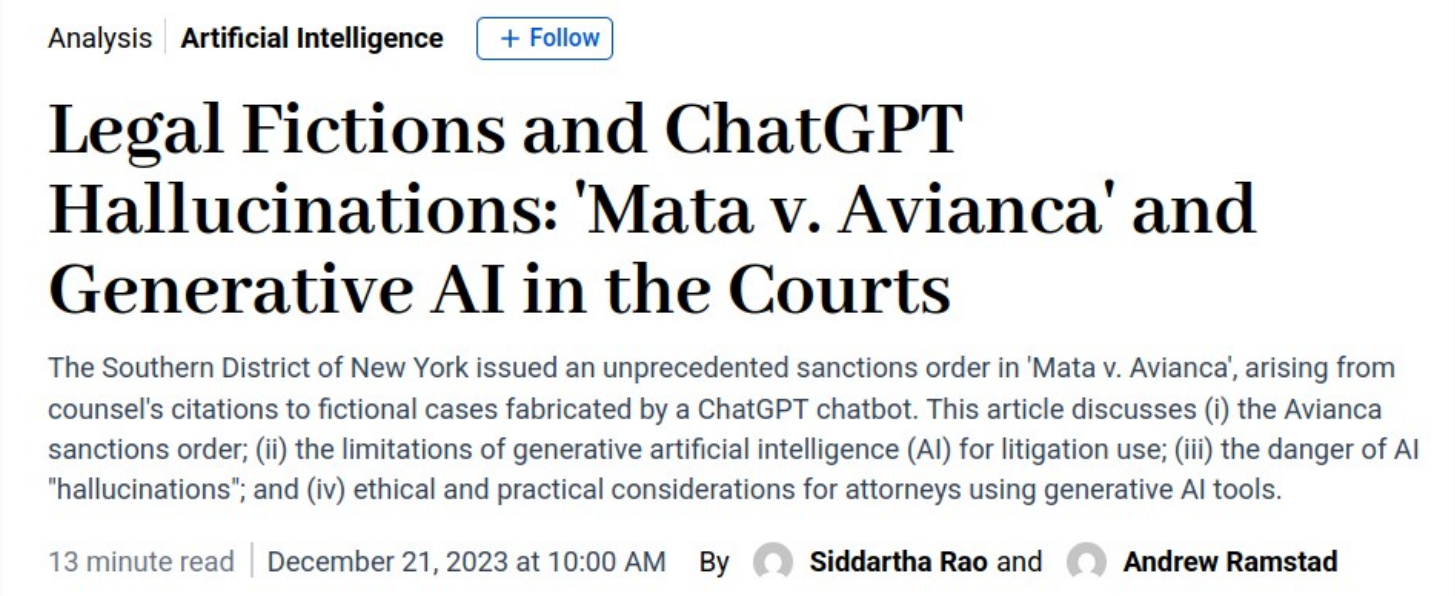
Four business controls

Give it the source: paste the policy text in

Fix the answers: a closed list, including "unknown"

Measure it: score it on examples with known answers

Human check where one error is expensive



Analysis | **Artificial Intelligence** + Follow

Legal Fictions and ChatGPT Hallucinations: 'Mata v. Avianca' and Generative AI in the Courts

The Southern District of New York issued an unprecedented sanctions order in 'Mata v. Avianca', arising from counsel's citations to fictional cases fabricated by a ChatGPT chatbot. This article discusses (i) the Avianca sanctions order; (ii) the limitations of generative artificial intelligence (AI) for litigation use; (iii) the danger of AI "hallucinations"; and (iv) ethical and practical considerations for attorneys using generative AI tools.

13 minute read | December 21, 2023 at 10:00 AM By **Siddartha Rao** and **Andrew Ramstad**

From chat to component: text in, a table row out

CUSTOMER EMAIL

“Third month in a row my bill is higher than my contract price. If this is not fixed by Friday I am moving to another operator.”



LLM OUTPUT (JSON)

```
{
"topic": "billing",
"sentiment": "negative",
"churn_risk": "high"
}
```

- **Countable:** 10,000 emails become a table you can filter and chart
- **Routable:** “high” goes to the retention team today
- **Testable:** compare each field with a human label

A business prompt has five parts

Part	In today's lab
1. Role	You sort customer emails for MegaTelCo, a telecom company (our example case since week 2).
2. Task	Read the email and label its topic, sentiment and churn risk.
3. Rules	topic is one of: billing, network, contract, technical, other. If unsure, choose "other".
4. Examples	Two or three emails with the correct labels ("few-shot").
5. Output format	Answer only with JSON: {"topic": ..., "sentiment": ..., "churn_risk": ...}

Rules and examples matter most. Write down what "high churn risk" means for your business, or the model will guess for you.

Huyen, AI Engineering, ch. 5 (prompt engineering).

Before you paste customer data into an AI tool

Free tiers are paid with your data. Google's pricing page says Gemini free-tier content is **used to improve its products**.

Today's lab uses invented emails only. Be careful when you feed sensitive data to these models!!!

With real customer data

1. Paid tier with a data-processing contract
2. Remove names, phone numbers, account IDs
3. Send only the fields the task needs
4. Check the rules: General Data Protection Regulation (GDPR) and Local laws always.

Source: ai.google.dev/gemini-api/docs/pricing, checked Nov 2026.

AI in customer service

Automate, assist or analyse, and what the evidence says



Three jobs for AI in customer service

Automate (AI Directly)

The AI answers the customer directly.

Opening hours, order status, password resets, routing emails

Errors reach the customer without directly!

Assist (Person Between)

The AI helps an agent, who decides.

Draft replies, suggested answers, call summaries, policy lookup

A person catches errors before it reaches customers

Analyse (AI far)

The AI reads everything; managers act.

Complaint topics, sentiment trends, churn signals in text

Errors average out, person decides

Rule of thumb: the closer the AI is to the customer, the higher the cost of a mistake.

Evidence: an AI assistant helped beginners most

SP500 Company: An AI assistant trained on the chats of a company's best support agents was given to 5,179 agents (outsourced people). On average they solved 14% more problems per hour, and beginners improved by 34%, reaching in 2 months what normally takes 6. The experts gained almost nothing. The AI acted as a way to pass the best agents' know-how to the newest ones. Customers were happier, and new staff were less likely to quit. Source: Brynjolfsson, Li & Raymond (2025), Generative AI at Work, Quarterly Journal of Economics.

+14%

issues resolved per hour, on average

+34%

for new and less-skilled agents

≈ 0

for the most experienced agents

What is the problem with this setup?

Klarna: from “700 agents” to hiring humans again

Klarna is a global financial technology company best known for its "Buy Now, Pay Later" (BNPL) service. It allows shoppers to purchase items immediately and split the cost into interest-free installments over time, acting as an alternative to traditional credit cards.

February 2024 · automate

Klarna’s AI assistant handled **2.3 million chats** in its first month, two-thirds of all customer-service chats.

The company said it did the work of **700 full-time agents**.

May 2025 · back to assist

The CEO said cost had weighed too heavily in the decision and **quality had fallen**.

Klarna started recruiting human agents again and promised customers they can **always reach a person**.

Lesson: they measured the cost per chat. They didn’t price the cost of a bad chat: an unsolved problem, an angry customer, a lost account.

Sources: Klarna press release, Feb 2024; CEO interview with Bloomberg, reported May 2025.

Routing an email (Remember last week CM)

The AI flags an email as high churn risk. Do we send it to the retention team?

	Really at risk	Not at risk
Send to retention	**+€57** ¼ saved × €240 − €3	**−€3** agent time wasted
Normal queue	**€0**	**€0**

A missed at-risk customer costs **19 times** more than a wasted escalation.

Break-even: escalate if the chance of risk is above **5%**.

So reward the AI for catching at-risk emails, even with some false alarms.

Activity 2: Automate, assist, or keep human?

What would you do with each of these agent?

1. “What are your shop opening hours?”

2. Replies to billing complaints

3. “Do I qualify for a refund?”

4. Monthly report on themes in 5,000 call transcripts

5. Closing the account of a customer who has died, with the family

6. Spotting customers who threaten to leave

Can we trust AI?

Evaluate it like a model; own it like an employee

Evaluate an LLM like any model from week 3

1

Label by hand

30 or more real examples, labelled by people who know the business

2

Set a baseline

Simple keyword rules: “cancel”, “switch”, “bill”

3

Compare

Accuracy and the confusion matrix, for the AI and the baseline

4

Price the errors

Euros per 1,000 emails, using the routing matrix

Today’s scoreboard	Topic accuracy	At-risk emails caught	€ per 1,000 emails
Keyword rules	__ %	__ of __	€ __
LLM, prompt v1	__ %	__ of __	€ __
LLM, prompt v2 (rules + examples)	__ %	__ of __	€ __

Mini-case: Moffatt v. Air Canada (2024)

The Incident: In November 2022, passenger Jake Moffatt needed to book an urgent flight following the death of his grandmother.

The Chatbot Error: He used the Air Canada website chatbot to ask about bereavement rates. The tool incorrectly told him he could book a full-fare ticket and apply for a retroactive refund within 90 days.

The Denial: When Moffatt applied for the discount after his trip, the airline denied the refund, stating that bereavement fares could not be applied retroactively and had to be requested before travel

THE CHATBOT SAID

After his grandmother died, Jake Moffatt asked about bereavement fares. He could buy now and claim the discount **within 90 days**.

THE POLICY SAID

The bereavement discount cannot be claimed after travel. Air Canada argued the chatbot was a **separate legal entity**, responsible for its own actions.

THE TRIBUNAL SAID

The chatbot is part of Air Canada's website. The airline is responsible for it. Pay about **C$812**.

Which control mechanism would have prevented this? Was this chatbot automating, or should it have been assisting?

Things to remember

1. A model much better than the baseline? Suspect leakage. Apply the time test.
2. An LLM is a component: text in, a table row out, priced per token.
3. Fluent is not true: give it the source, fix the answers, check it.
4. Judge it like any model: labelled examples, a baseline, the cost of each error.
5. The company owns every word its chatbot says.